Introduction to Representation Theory

Feb. 9 Week 1-2

Recall G = group $K \in \{\mathbb{R}, \mathbb{C}\}$

V = finite dimensional K-vector space

A linear representation of G on V is a <u>group homomorphism</u> $\pi : G \to GL_n(K)$ for some $n \in \mathbb{N}$

Saw: Have bijective correspondence

$$\{\text{eq cl of linear reps}\} \leftrightarrow \{\text{eq. of matrix reps}\}.$$

<u>Defn</u>: Suppose (π, V) of G. Then a subspace $W \subseteq V$ is called <u>invariant</u> if $\forall g \in G$ $\pi(g) W \subseteq W$.

If $W \subseteq V$ is an invariant subspace, we define (π_W, W) of G on W by $\pi_W(g) = \pi(g)|_W : W \to W$

We call (π_W, W) a sub-representation of π.

<u>Note</u> Sometimes if we call a repn (σ, U) of G a subrep'n of (π, V) if $\exists$ invariant $W \subseteq V$ s.t. (π_W, W)

<u>Example</u>: Recall the rep'n $\pi : S_n \to GL_n(K) = GL(K^n)$

$$\pi(\sigma) e_j = e_{\sigma(j)}$$

Then $W = K(e_1 + \cdots + e_n) \subseteq K^n$ is invariant

In fact (π_W, W) is the <u>trivial</u> representation. i.e. $\pi_W(\sigma) = id_W$ $\forall \sigma \in S_n$

<u>Remark</u>: If $W_1, W_2 \subseteq V$ are two invariant subspaces then $W_1 \cap W_2$ and $W_1 + W_2$ are also invariant.

<u>Def'n/Lemma</u> Suppose (π, V) is a rep'n of G $W \subseteq V$ invariant subspaces Then we define the quotient rep'n of G on V/W by

$$\pi_{V/W}(g)(v+W) = \pi(g)v + W.$$

This is a well-defined rep'n of G on V/W. Moreover if $V = W \oplus \widetilde{W}$ with $\widetilde{W} \subseteq V$ also invariant, then

$$(\pi_{V/W}, V/W) \simeq (\pi_{\widetilde{W}}, \widetilde{W})$$

<u>Proof</u>: $\pi_{V/W}$ is well-defined. Let $v_1 + W, v_2 + W \in V/W$, s.t. $v_1 + W = v_2 + W$. Then

$$\pi(g)v_1 + W - (\pi(g)v_2 + W) = \pi(g)\underbrace{(v_1 - v_2)}_{\in W} + W = 0 + W$$

$\pi_{V/W}$ is a rep'n.

$$\begin{aligned}
\pi_{V/W}(gh)(v+W) &= \pi(gh)v + W \\
&= \pi(g)\pi(h)v + W \\
&= \pi_{V/W}(g)(\pi(h)v + W) \\
&= \pi_{V/W}(g) \circ \pi_{V/W}(h)(v+W)
\end{aligned}$$

Now suppose $V = W \oplus \tilde{W}$ with $\tilde{W} \subseteq V$ invariant

Then $\Phi : \tilde{W} \longrightarrow V/W \quad \tilde{w} \longmapsto \tilde{w} + W$ is a vector space isomorphism and gives equivalence

$$(\pi_{V/W}, V/W) \simeq (\pi_{\tilde{W}}, \tilde{W}) \qquad \square$$

<u>Def'n</u> A repn (π, V) of G is called <u>irreducible</u> if the only invariant subspaces of V are $\{0\}$ and V.

Example 1) 1-dim'l representations are always irreducible.

2) Recall the rep'n $\sigma : \mathbb{R} \longrightarrow GL_2(\mathbb{R})$

$$t \longmapsto \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$$

σ is irreducible

If σ were not irreducible $\exists\, v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in \mathbb{R}^2 \setminus \{0\}$ s.t. $\mathbb{R}_v$ is invariant But $\sigma(\frac{\pi}{2})\left(\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}\right) = \begin{bmatrix} -v_2 \\ v_1 \end{bmatrix} \notin \mathbb{R}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$

3) $G = GL(V)$ then $\pi : G \to GL(V)$

$$g \longmapsto g$$

is irreducible.

4) Back to $\pi : S_n \longrightarrow GL(K^n) \quad \pi(\sigma)(e_j) = e_{\sigma(j)}$

$W = K(e_1 + \cdots + e_n)$ is an invariant subspace and (π_W, W) is irreducible.

· Define $U = \{ a_1e_1 + \cdots + a_ne_n \in K^n \mid \underline{\sum_{j=1}^{n} a_j = 0} \} \subseteq K^n$.

Then clearly $K^n = W \oplus U$ and U is an invariant subspaces.

$\rightsquigarrow (\pi_{K^n/W}, K^n/W) \simeq (\pi_U, U)$

In fact (π_U, U) is irreducible.

$$\boxed{\begin{array}{l}\underline{\text{Note}} \text{ If } \sigma_1, \sigma_2 \text{ are two rep'ns of } G \text{ s.t } \sigma_1 \simeq \sigma_2 \\ \text{Then } \sigma_1 \text{ is irreducible} \iff \sigma_2 \text{ is irreducible}\end{array}}$$

<u>Note</u> $e_1 - e_2, e_1 - e_3, \ldots, e_1 - e_n$ is a basis for U.

Let $\tilde{U} \subseteq U$ be an invariant subspace, $\tilde{U} \neq 0$. Let

$u = a_1e_1 + \cdots + a_ne_n \in \tilde{U}$. s.t. $\exists\, i, j$ s.t. $a_i \neq a_j$

Then $\pi((ij))\, u - u = a_je_i + a_ie_j - (a_ie_i + a_je_j)$

$\tilde{U} \ni$

$$= \underbrace{(a_j - a_i)}_{\neq 0}(e_i - e_j)$$

$\Rightarrow e_i - e_j \in \tilde{U}$

If $j \neq 1$ (if $j = 1$ consider $-(e_i - e_j) = e_j - e_i$ and swit rules of i & j)

$\Rightarrow e_1 - e_j = \pi((i\,1))(e_i - e_j) \in \tilde{U}$

$\Rightarrow \forall k \in \{2, \ldots, n\}$ $\underbrace{e_1 - e_k}_{\text{basis element}} = \pi(kj)(e_i - e_j) \in \tilde{U}$

$\Rightarrow \tilde{U} = U$ $\square$

<u>Def'n</u> A rep'n (π, V) of G is called completely reducible if $\forall$ <u>invariant subspace</u> $W \subseteq V$, $\exists$ <u>inv subspace</u> $\widetilde{W} \subseteq V$ s.t. $V = W \oplus \widetilde{W}$

<u>Example</u> 1) Every <u>irreducible</u> representation is <u>completely reducible</u>

2) $\pi : S_n \longrightarrow GL(K^n)$ $\pi(\sigma)(e_j) = e_{\sigma j}$, is completely reducible.

3) $\pi : \mathbb{R} \longrightarrow GL_2(\mathbb{R})$ $\pi(t) = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix} = e^{t\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}$

π is <u>not</u> irreducible because $\mathbb{R}e_1$ is invariant.

But π is <u>neither</u> completely reducible.

For every $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in \mathbb{R}^2$, $v_2 \neq 0$

$$\pi(t)(v) = \begin{bmatrix} v_1 + t v_2 \\ v_2 \end{bmatrix}$$

If $\mathbb{R}v$ were invariant, then

$$\mathbb{R}v \ni \pi(t)v - v = \begin{bmatrix} t v_2 \\ 0 \end{bmatrix} \notin \mathbb{R}v \text{ a contradiction}$$

$\Rightarrow \pi$ is not completely reducible.

$$\boxed{\text{<u>Maschke's Theorem</u> Every real or complex finite dimensional representation of a finite group is completely reducible.}}$$

To be proved next week

<u>Proposition 1</u> If (π, V) is a completely reducible rep'n of G and $W \subseteq V$ an invariant subspace, (π_W, W) is also completely reducible.

<u>Proof</u>. Suppose $U \subseteq W$ is invariant for π_W, then U is also invariant for π.

$\xRightarrow[\text{reducible}]{\pi \text{ completely}}$ $\exists$ inv $\tilde{U} \subseteq V$ s.t. $V = U \oplus \tilde{U}$

Then $\tilde{U} \cap W$ is invariant and $W = U \oplus (\tilde{U} \cap W)$

so (π_W, W) is also completely reducible. $\square$

<u>Proposition 2</u> If (π, V) is a completely reducible rep'n of G, then $\exists$ minimal invariant subspaces $W_1, \cdots W_\ell \subseteq V$ s.t. $V = W_1 \oplus \cdots \oplus W_\ell$.

<u>Note</u>. An invariant subspace $W \subseteq V$ is <u>minimal</u> $\Leftrightarrow$ <u>(π_W, W) is irreducible</u>

$\boxed{\text{Recall we always assume that } V \text{ is finite dim'l}}$

<u>Proof of Prop 2</u>. If π is irred, we are done

$\left.\begin{array}{l}\text{If } \pi \text{ is not irreducible then} \\ \exists \text{ invariant subspace } 0 \neq W \subsetneq V \\ \text{and since } \pi \text{ is completely reducible, } \exists\ \tilde{W} \\ \text{s.t. } V = W \oplus \tilde{W}\end{array}\right\} (*)$

Prop 1 $\Rightarrow (\pi_W, W)$, $(\pi_{\tilde{W}}, \tilde{W})$ are completely reducible

Apply $\circledast$ to (π_W, W) and $(\pi_{\tilde{W}}, \tilde{W})$ if they are not irreducible. This terminates after finitely many steps because $\dim V < \infty$ $\square$

<u>Theorem</u> Let (π, V) be a rep'n of G

Suppose $W_1, \dots, W_\ell \subseteq V$ are minimal invariant subspaces such that

$\circledast\circledast$ $\quad V = W_1 + \dots + W_\ell$ (not necessarily direct)

Then (π, V) is completely reducible and $\forall$ invariant subspace $U \subseteq V$

$\exists \{i_1, \dots, i_r\} \subseteq \{1, \dots, \ell\}$ s.t

might be empty

$$V = U \oplus W_{i_1} \oplus \dots \oplus W_{i_r} \qquad \circledast$$

Proof: It will suffices to prove $\circledast$. Suppose $U \subseteq V$ is an invariant subspace.

Let $I = \{i_1, \dots, i_r\} \subseteq \{1, \dots, \ell\}$ maximal s.t

$U, W_{i_1}, \dots, W_{i_r}$ is linearly independent

i.e. $U + W_{i_1} + \dots + W_{i_r} = U \oplus W_{i_1} \oplus \dots \oplus W_{i_r} =: X$

I might be empty!

If $X = V$ we are done.

Suppose $X \subsetneq V$

$\overset{(**)}{\Longrightarrow}$ $j \in \{1, \dots, l\} \setminus I$ s.t. $W_j \subseteq X$

W_j & X are invariant $\Rightarrow$ $W_j \cap X \subseteq W_j$ invariant

$\overset{W_j \text{ minimal}}{\Longrightarrow}$ $W_j \cap X = \{0\}$ or $\underbrace{W_j \cap X = W_j}_{\substack{\text{not possible} \\ \text{since } W_j \not\subseteq X}}$

$\Rightarrow$ $W_j \cap X = \{0\}$

$\Rightarrow$ $U, W_{i_1} \dots W_{i_r}, W_l$ is linearly independent

but this contradicts the fact that I is maximal

$\Rightarrow$ $X = V$. $\square$

<u>Example</u> Let $A \in M_n(\mathbb{C})$

$$\pi_A : \mathbb{R} \to GL_n(\mathbb{C}) \quad \pi_A(t) = e^{tA}$$

Claim: π_A is completely reducible

$\Leftrightarrow$ A is diagonalizable.

Proof: "$\Leftarrow$" A is diagonalizable $\Rightarrow$ $\exists B \in GL_n(\mathbb{C})$

s.t. $B^{-1}AB = \begin{bmatrix} a_1 & & \\ & \ddots & \\ & & a_n \end{bmatrix} = D$

Know $B^{-1} e^{tA} B = e^{t B^{-1}AB}$

$\Rightarrow$ $\pi_A \simeq \pi_D$

π_D is completely reducible because $\mathbb{C}^n = \bigoplus_{j=1}^{n} \mathbb{C}e_j$

and $\mathbb{C}e_j$ is invariant under π_D, because

$$\pi_D(t)(e_j) = e^{ta_j} e_j \in \mathbb{C}e_j$$

"$\Rightarrow$" Assume A is not diagonalizable.

Wlog A is in Jordan normal form:

$$A = \begin{bmatrix} a_1 & 1 & & \\ 0 & a_1 & & 0 \\ & & \ddots & \\ 0 & & & \ddots \end{bmatrix}$$

Then $\mathbb{C}e_1$ is an invariant subspace of $\mathbb{C}^n$ for π_A but it does not have an invariant complement

$\square$